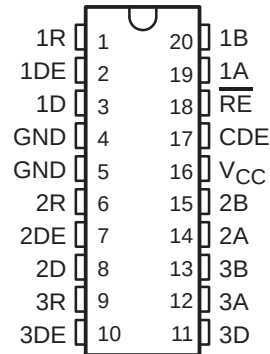


SN75ALS171, SN75ALS171A TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

SLLS056D – AUGUST 1987 – REVISED SEPTEMBER 1995

- Three Bidirectional Transceivers
- Driver Meets or Exceeds the Requirements of ANSI EIA/TIA-422-B and RS-485 and ITU Recommendation V.11
- Two Skew Limits Available
- Designed to Operate Up to 20 Million Data Transfers per Second (FAST-20 SCSI)
- High-Speed Advanced Low-Power Schottky Circuitry
- Low Pulse Skew . . . 5 ns Max
- Designed for Multipoint Transmission on Long Bus Lines in Noisy Environments
- Features Independent Driver Enables and Combined Receiver Enables
- Wide Positive and Negative Input/Output Bus Voltages Ranges
- Driver Output Capacity . . . ± 60 mA
- Thermal Shutdown Protection
- Driver Positive- and Negative-Current Limiting
- Receiver Input Impedances . . . 12 k Ω Min
- Receiver Input Sensitivity . . . ± 300 mV Max
- Receiver Input Hysteresis . . . 60 mV Typ
- Operates From a Single 5-V Supply
- Glitch-Free Power-Up and Power-Down Protection

DW OR J PACKAGE
(TOP VIEW)



description

The SN75ALS171 and the SN75ALS171A triple differential bus transceivers are monolithic integrated circuits designed for bidirectional data communication on multipoint bus transmission lines. They are designed for balanced transmission lines, and each driver meets ANSI Standards EIA/TIA-422-B and RS-485 and both the drivers and receivers meet ITU Recommendation V.11. The SN75ALS171A is designed for FAST-20 SCSI and can transmit or receive data pulses as short as 30 ns with a maximum skew of 5 ns.

The SN75ALS171 and the SN75ALS171A operate from a single 5-V power supply. The drivers and receivers have individual active-high and active-low enables, respectively, which can be externally connected together to function as a direction control. The driver differential output and the receiver differential input pairs are connected internally to form differential input/output (I/O) bus ports that are designed to offer minimum loading to the bus when the driver is disabled or V_{CC} is at 0 V. These ports feature wide positive and negative common-mode voltage ranges making the device suitable for party-line applications.

The SN75ALS171 and the SN75ALS171A are characterized for operation from 0°C to 70°C.



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

 **TEXAS
INSTRUMENTS**

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SN75ALS171, SN75ALS171A

TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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Function Tables

EACH DRIVER

INPUT D	ENABLES		OUTPUTS	
	DE	CDE	A	B
H	H	H	H	L
L	H	H	L	H
X	L	X	Z	Z
X	X	L	Z	Z

EACH RECEIVER

DIFFERENTIAL INPUTS A – B	ENABLE $\overline{RE}$	OUTPUT R
$V_{ID} \geq 0.3 \text{ V}$	L	H
$-0.3 \text{ V} < V_{ID} < 0.3 \text{ V}$	L	?
$V_{ID} \leq -0.3 \text{ V}$	L	L
X	H	Z
Open	L	H

H = high level, L = low level, ? = indeterminate, X = irrelevant,
Z = high impedance (off)

AVAILABLE OPTIONS

SKEW LIMIT	PART NUMBER	
10 ns	SN75ALS171DW	SN75ALS171J
5 ns	SN75ALS171ADW	

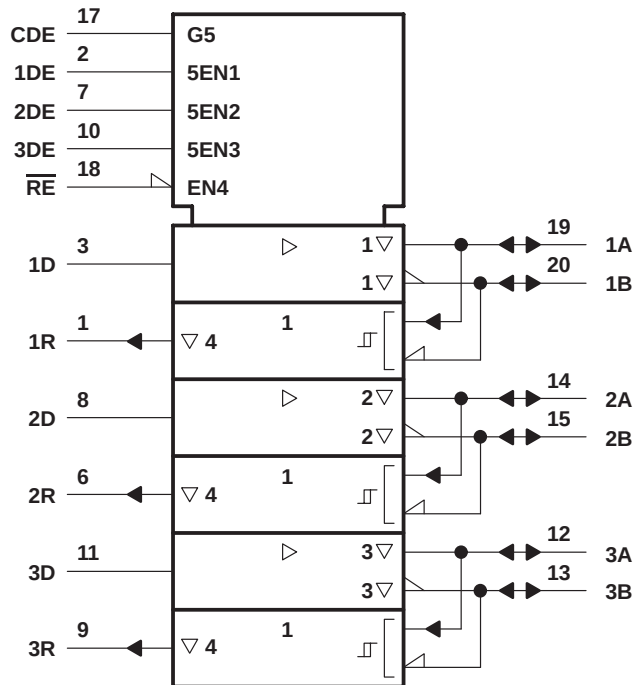


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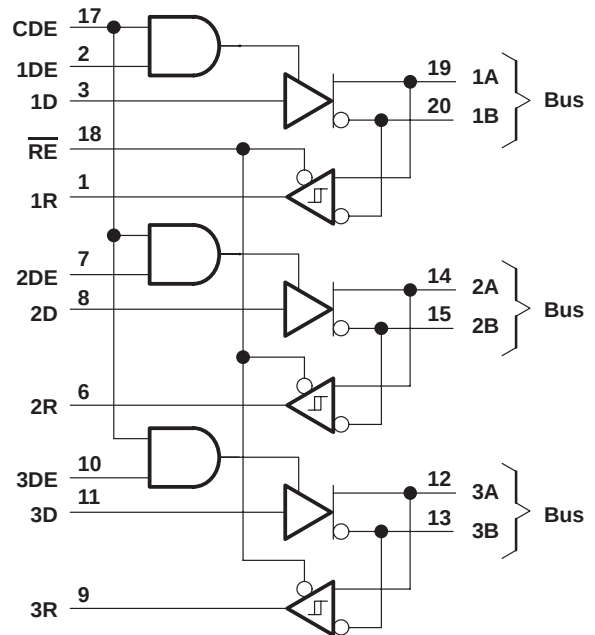
SN75ALS171, SN75ALS171A TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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logic symbol†

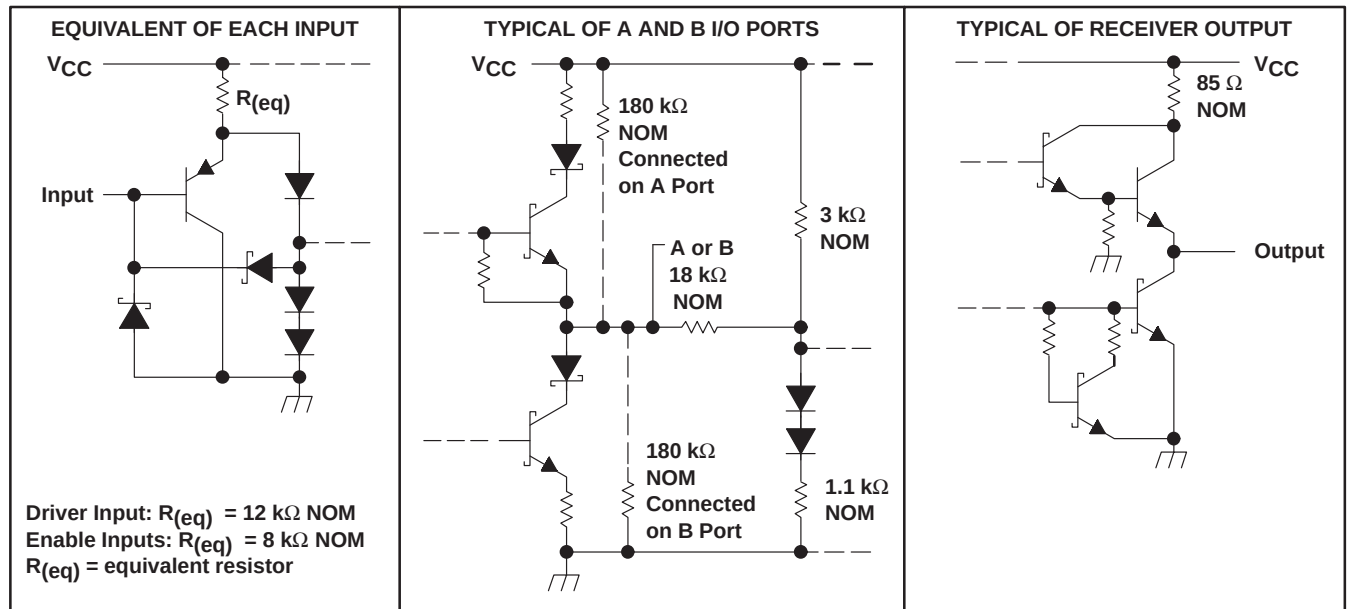


logic diagram (positive logic)



† This symbol is in accordance with ANSI/IEEE Std 91-1984 and IEC Publication 617-12.

schematics of inputs and outputs



SN75ALS171, SN75ALS171A

TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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absolute maximum ratings over operating free-air temperature range (unless otherwise noted)[†]

Supply voltage, V_{CC} (see Note 1)	7 V
Voltage range at any bus terminal	–7 V to 12 V
Enable input voltage, V_I	7 V
Continuous total power dissipation	See Dissipation Rating Table
Operating free-air temperature range, T_A	0°C to 70°C
Storage temperature range, T_{stg}	–65°C to 150°C
Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds: DW package	260°C
Lead temperature 1,6 mm (1/16 inch) from case for 60 seconds: J package	300°C

[†] Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

NOTE 1: All voltage values, except differential I/O bus voltage, are with respect to network ground terminal.

DISSIPATION RATING TABLE

PACKAGE	$T_A \leq 25^\circ\text{C}$ POWER RATING	DERATING FACTOR ABOVE $T_A = 25^\circ\text{C}$	$T_A = 70^\circ\text{C}$ POWER RATING
DW	1125 mW	9.0 mW/°C	720 mW
J	1025 mW	8.2 mW/°C	656 mW

recommended operating conditions

		MIN	NOM	MAX	UNIT
Supply voltage, V_{CC}		4.75	5	5.25	V
Voltage at any bus terminal (separately or common mode), V_I or V_{IC}		–7		12	V
High-level input voltage, V_{IH}	D, CDE, DE, and $\overline{RE}$	2			V
Low-level input voltage, V_{IL}	D, CDE, DE, and $\overline{RE}$			0.8	V
Differential input voltage, V_{ID} (see Note 2)				±12	V
High-level output current, I_{OH}	Driver			–60	mA
	Receiver			–400	μA
Low-level output current, I_{OL}	Driver			60	mA
	Receiver			8	mA
Operating free-air temperature, T_A		0		70	°C

NOTE 2: Differential-input/output bus voltage is measured at the noninverting terminal A with respect to the inverting terminal B.



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SN75ALS171, SN75ALS171A

TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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DRIVER SECTION

electrical characteristics over recommended ranges of supply voltage and operating free-air temperature (unless otherwise noted)

PARAMETER		TEST CONDITIONS†		MIN	TYP‡	MAX	UNIT
V _{IK}	Input clamp voltage	I _I = −18 mA				−1.5	V
V _O	Output voltage	I _O = 0		0		6	V
V _{OH}	High-level output voltage	V _{CC} = 4.75 V, V _{IL} = 0.8 V,	V _{IH} = 2 V, I _{OH} = −55 mA	2.7			V
V _{OL}	Low-level output voltage	V _{CC} = 4.75 V, V _{IL} = 0.8 V,	V _{IH} = 2 V, I _{OL} = 55 mA			1.7	V
V _{OD1}	Differential output voltage	I _O = 0		1.5		6	V
V _{OD2}	Differential output voltage	R _L = 100 Ω,	See Figure 1	1/2V _{OD1} or 2§	2.5	5	V
		R _L = 54 Ω,	See Figure 1	1.5	2.5	5	
V _{OD3}	Differential output voltage	V _{test} = −7 V to 12 V, See Figure 2		1.5		5	V
Δ V _{OD}	Change in magnitude of differential output voltage¶	R _L = 54 Ω or 100 Ω, See Figure 1		±0.2			V
V _{OC}	Common-mode output voltage			3			V
				−1			
Δ V _{OC}	Change in magnitude of common-mode output voltage¶			±0.2			V
I _O	Output current	Output disabled, See Note 3	V _O = 12 V	1			mA
			V _O = −7 V	−0.8			
I _{IH}	High-level enable-input current	D and DE	V _{IH} = 2.7 V	20			μA
		CDE		60			
I _{IL}	Low-level enable-input current	D and DE	V _{IL} = 0.4 V	−100			
		CDE		−900			
I _{OS}	Short-circuit output current	V _O = −6 V		−250			mA
		V _O = 0		−150			
		V _O = V _{CC}		250			
		V _O = 8 V		250			
I _{CC}	Supply current	No load	Outputs enabled	69	90	mA	
			Outputs disabled	57	78		

† The power-off measurement in ANSI Standard EIA/TIA-422-B applies to disabled outputs only and is not applied to combined inputs and outputs.

‡ All typical values are at $V_{CC} = 5$ V and $T_A = 25^\circ\text{C}$.

§ The minimum V_{OD2} with 100- Ω load is either $1/2 V_{OD2}$ or 2 V, whichever is greater.

¶ $\Delta|V_{OD}|$ and $\Delta|V_{OC}|$ are the changes in magnitude of V_{OD} and V_{OC} , respectively, that occur when the input is changed from a high level to a low level.

NOTE 3: This applies for both power on and off; refer to EIA Standard RS-485 for exact conditions. The EIA/TIA-422-B limit does not apply for a combined driver and receiver terminal.



SN75ALS171, SN75ALS171A

TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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switching characteristics over recommended ranges of supply voltage and operating free-air temperature (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP [†]	MAX	UNIT
t _d (OD)	Differential output delay time	ALS171	R _L = 54 Ω, See Figure 3,	3		13	ns
		ALS171A	C _L = 50 pF	6		11	
		ALS171	R _{L1} = R _{L3} = 165 Ω, V _{TERM} = 5 V, See Figure 6	3		13	
		ALS171A	C _L = 60 pF, R _{L2} = 75 Ω,	6		11	
t _{sk} (p)	Pulse skew [‡]	R _L = 54 Ω, C _L = 50 pF, See Figure 3		1		5	ns
		R _{L1} = R _{L3} = 165 Ω, R _{L2} = 75 Ω, C _L = 60 pF, See Figure 6		1		5	ns
t _{sk} (lim)	Skew limit [§]	ALS171	R _L = 54 Ω, C _L = 50 pF,	10			ns
		ALS171A	See Figure 3	5			
		ALS171	R _{L1} = R _{L3} = 165 Ω, R _{L2} = 75 Ω, C _L = 60 pF, See Figure 6	10			
		ALS171A		5			
t _t (OD)	Differential-output transition time	R _L = 54 Ω, C _L = 50 pF, See Figure 3		3	8	13	ns
		R _{L1} = R _{L3} = 165 Ω, R _{L2} = 75 Ω, C _L = 60 pF, V _{TERM} = 5 V, See Figure 6		3	8	13	
t _{pZH}	Output enable time to high level	R _L = 110 Ω, See Figure 4		30		50	ns
t _{pZL}	Output enable time to low level	R _L = 110 Ω, See Figure 5		30		50	ns
t _{PHZ}	Output disable time from high level	R _L = 110 Ω, See Figure 4		3	8	13	ns
t _{PLZ}	Output disable time from low level	R _L = 110 Ω, See Figure 5		3	8	13	ns
t _{pDE}	Differential-output enable time	R _{L1} = R _{L3} = 165 Ω, R _{L2} = 75 Ω,		8	30	45	ns
t _{pDZ}	Differential-output disable time	C _L = 60 pF, See Figure 7		5	10	45	ns

[†] All typical values are at $V_{CC} = 5 \text{ V}$ and $T_A = 25^\circ\text{C}$.

[‡] Pulse skew is defined as the $|t_{d(ODH)} - t_{d(ODL)}|$ of each channel.

[§] Skew limit is the maximum difference in propagation delay times between any two channels of one device and between any two devices. This parameter is applicable at one V_{CC} and operating temperature within the recommended operating conditions.

SYMBOL EQUIVALENTS

DATA-SHEET PARAMETER	EIA/TIA-422-B	RS-485
V_O	V_{0a}, V_{0b}	V_{0a}, V_{0b}
$ V_{OD1} $	V_0	V_0
$ V_{OD2} $	$V_t (R_L = 100 \Omega)$	$V_t (R_L = 54 \Omega)$
$ V_{OD3} $		V_t (Test Termination Measurement 2)
V_{test}		V_{tst}
$\Delta V_{OD} $	$ V_t - \bar{V}_t $	$ V_t - \bar{V}_t $
V_{OC}	$ V_{os} $	$ V_{os} $
$\Delta V_{OC} $	$ V_{os} - \bar{V}_{os} $	$ V_{os} - \bar{V}_{os} $
I_{OS}	$ I_{sa} , I_{sb} $	
I_O	$ I_{xa} , I_{xb} $	I_{ia}, I_{ib}



RECEIVER SECTION

electrical characteristics over recommended ranges of common-mode input voltage, supply voltage, and operating free-air temperature (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP [†]	MAX	UNIT
V _{IT+}	Positive-going input threshold voltage	V _O = 2.7 V,	I _O = –0.4 mA			0.3	V
V _{IT–}	Negative-going input threshold voltage	V _O = 0.5 V,	I _O = 8 mA	–0.3 [‡]			V
V _{hys}	Hysteresis voltage (V _{IT+} – V _{IT–})				60		mV
V _{IK}	Enable-input clamp voltage	I _I = –18 mA				–1.5	V
V _{OH}	High-level output voltage	V _{ID} = 300 mV, See Figure 8	I _{OH} = –400 µA,	2.7			V
V _{OL}	Low-level output voltage	V _{ID} = –300 mV, See Figure 8	I _{OL} = 8 mA,			0.45	V
I _{OZ}	High-impedance-state output current	V _O = 0.4 V to 2.4 V				±20	µA
I _I	Line input current	Other input = 0 V, See Note 4	V _I = 12 V			1	mA
			V _I = –7 V			–0.8	
I _{IH}	High-level enable-input current	V _{IH} = 2.7 V				60	µA
I _{IL}	Low-level enable-input current	V _{IL} = 0.4 V				–300	µA
r _i	Input resistance			12			kΩ
I _{OS}	Short-circuit output current	V _{ID} = 300 mV,	V _O = 0	–15		–85	mA
I _{CC}	Supply current	No load	Outputs enabled		69	90	mA
			Outputs disabled		57	78	

[†] All typical values are at V_{CC} = 5 V and T_A = 25°C.

[‡] The algebraic convention, in which the less positive (more negative) limit is designated minimum, is used in this data sheet for common-mode input voltage and threshold voltage levels only.

NOTE 4: This applies for both power on and off; refer to EIA Standard RS-485 for exact conditions.

switching characteristics over recommended ranges of supply voltage and operating free-air temperature range

PARAMETER		TEST CONDITIONS	MIN	TYP [†]	MAX	UNIT
t _{PLH}	Propagation delay time, low- to high-level output	V _{ID} = −1.5 V to 1.5 V, C _L = 15 pF, T _A = 25°C, See Figure 9	9		19	ns
			11		16	
t _{PHL}	Propagation delay time, high- to low-level output		9		19	ns
			11		16	
t _{sk(p)}	Pulse skew [§]	V _{ID} = −1.5 V to 1.5 V, C _L = 15 pF, See Figure 9		2	5	ns
t _{sk(lim)}	Skew limit [¶]				10	ns
					5	
t _{PZH}	Output enable time to high level	C _L = 15 pF, See Figure 10		7	14	ns
t _{PZL}	Output enable time to low level			7	14	ns
t _{PHZ}	Output disable time from high level	C _L = 15 pF, See Figure 10		20	35	ns
t _{PLZ}	Output disable time from low level			8	17	ns

[†] All typical values are at V_{CC} = 5 V and T_A = 25°C.

[§] Pulse skew is defined as the |t_{PLH} – t_{PHL}| of each channel.

[¶] Skew limit is the maximum difference in propagation delay times between any two channels of one device and between any two devices. This parameter is applicable at one V_{CC} and operating temperature within the recommended operating conditions.

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PARAMETER MEASUREMENT INFORMATION

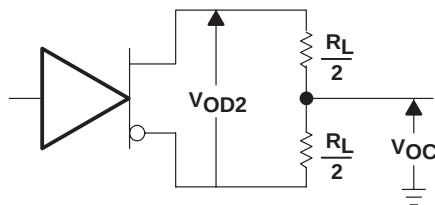


Figure 1. Driver V_{OD} and V_{OC}

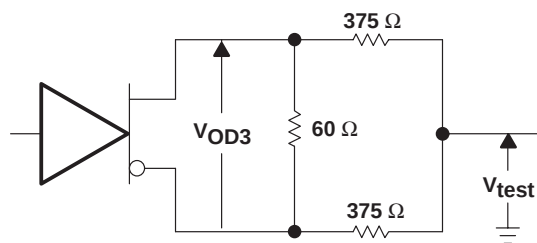
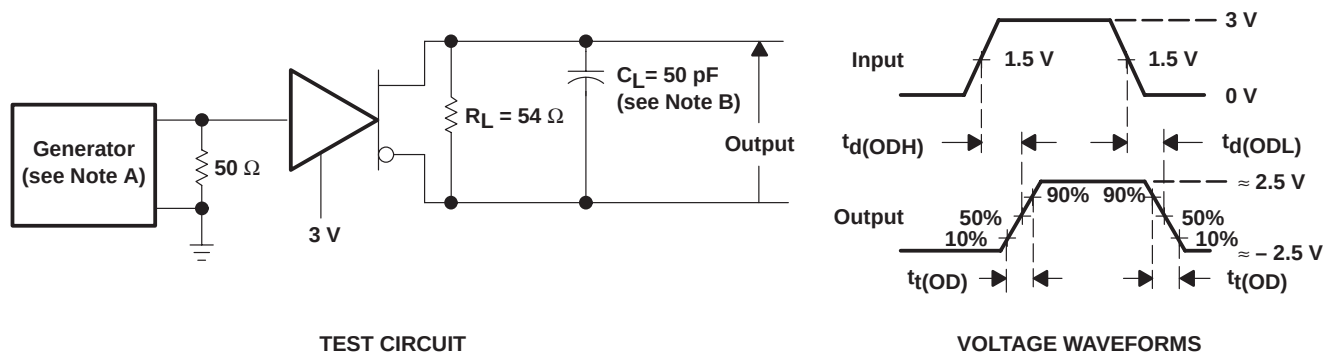


Figure 2. Driver V_{OD3}



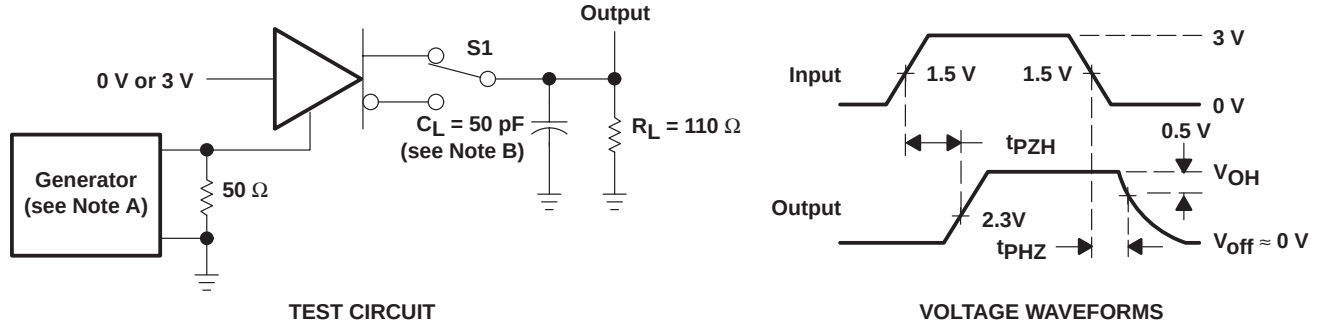
TEST CIRCUIT

VOLTAGE WAVEFORMS

- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $PRR \leq 1$ MHz, 50% duty cycle, $t_r \leq 6$ ns, $t_f \leq 6$ ns, $Z_O = 50 \Omega$.
B. C_L includes probe and jig capacitance.

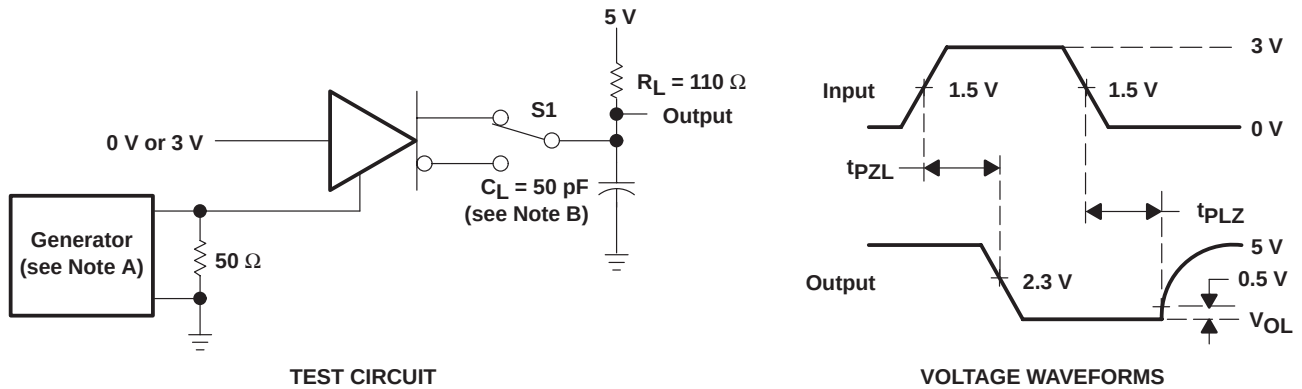
Figure 3. Driver Test Circuit and Voltage Waveforms

PARAMETER MEASUREMENT INFORMATION



- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $PRR \leq 1$ MHz, 50% duty cycle, $t_r \leq 6$ ns, $t_f \leq 6$ ns, $Z_O = 50 \Omega$.
 B. C_L includes probe and jig capacitance.

Figure 4. Driver Test Circuit and Voltage Waveforms



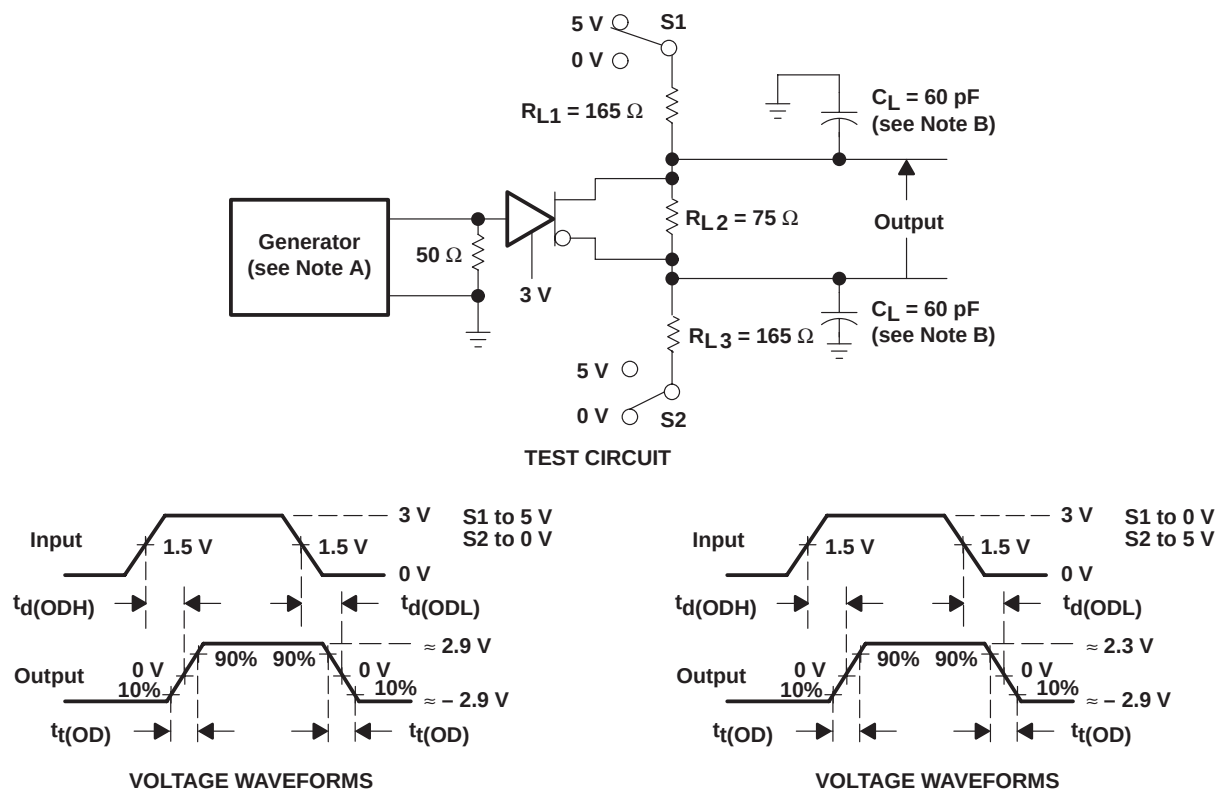
- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $PRR \leq 1$ MHz, 50% duty cycle, $t_r \leq 6$ ns, $t_f \leq 6$ ns, $Z_O = 50 \Omega$.
 B. C_L includes probe and jig capacitance.

Figure 5. Driver Test Circuit and Voltage Waveforms

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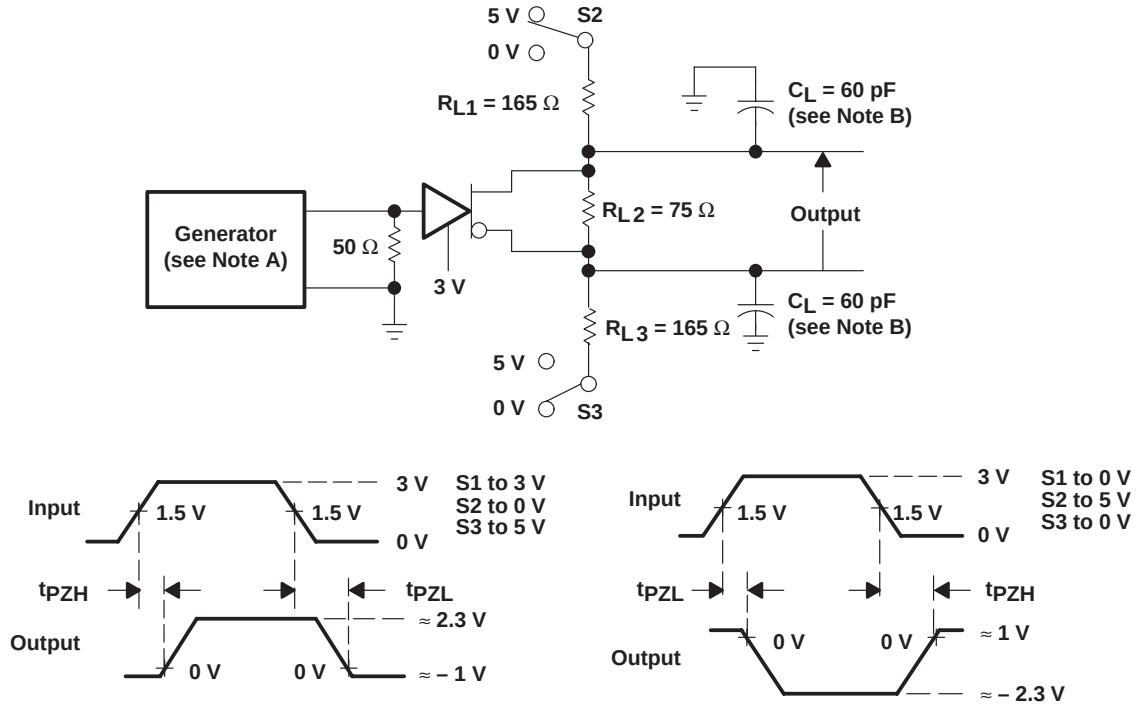
PARAMETER MEASUREMENT INFORMATION



- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $\text{PRR} \leq 1 \text{ MHz}$, 50% duty cycle, $t_r \leq 6 \text{ ns}$, $t_f \leq 6 \text{ ns}$, $Z_O = 50 \Omega$.
B. C_L includes probe and jig capacitance.

**Figure 6. Driver Test Circuit and Voltage Waveforms
With Double-Differential-SCSI Termination for the Load**

PARAMETER MEASUREMENT INFORMATION



- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $\text{PRR} \leq 1 \text{ MHz}$, 50% duty cycle, $t_r \leq 6 \text{ ns}$, $t_f \leq 6 \text{ ns}$, $Z_O = 50 \Omega$.
B. C_L includes probe and jig capacitance.

Figure 7. Driver Differential-Enable and Disable Times With a Double-SCSI Termination

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PARAMETER MEASUREMENT INFORMATION

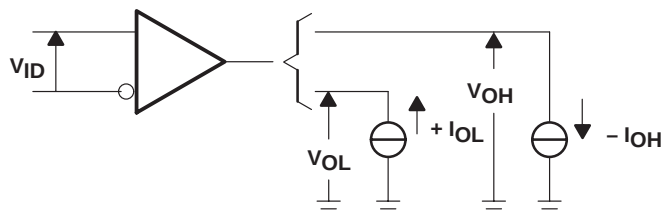
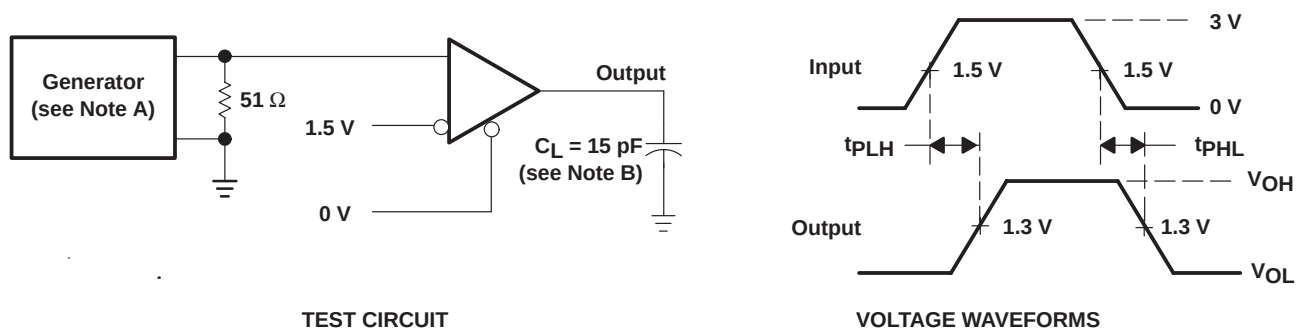


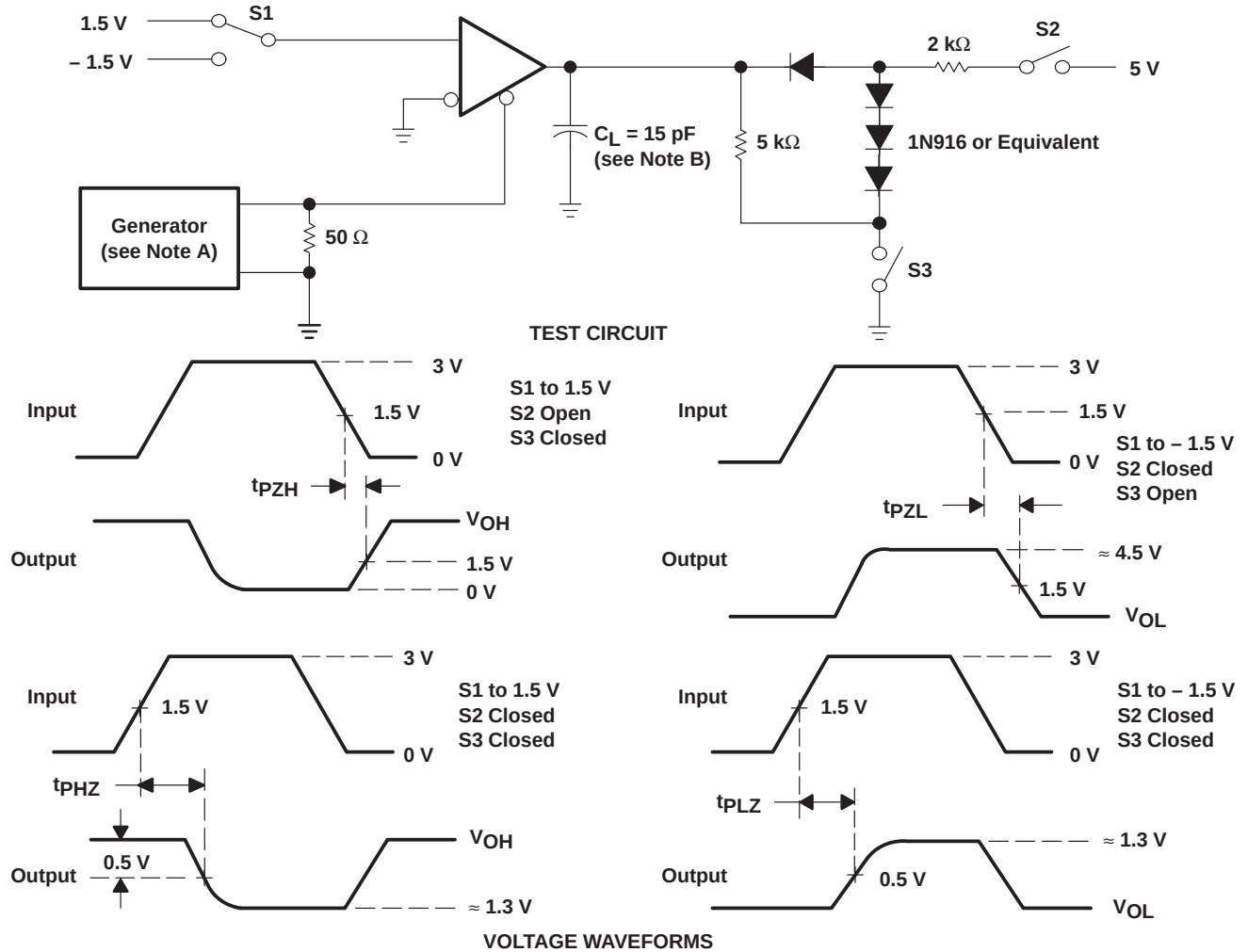
Figure 8. Receiver V_{OH} and V_{OL}



- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $PRR \leq 1 \text{ MHz}$, 50% duty cycle, $t_r \leq 6 \text{ ns}$, $t_f \leq 6 \text{ ns}$, $Z_O = 50 \Omega$.
B. C_L includes probe and jig capacitance.

Figure 9. Receiver Test Circuit and Voltage Waveforms

PARAMETER MEASUREMENT INFORMATION



- NOTES: A. The input pulse is supplied by a generator having the following characteristics: $PRR \leq 1$ MHz, 50% duty cycle, $t_r \leq 6$ ns, $t_f \leq 6$ ns, $Z_O = 50 \Omega$.
- B. C_L includes probe and jig capacitance.

Figure 10. Receiver Test Circuit and Voltage Waveforms

SN75ALS171, SN75ALS171A
TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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TYPICAL CHARACTERISTICS

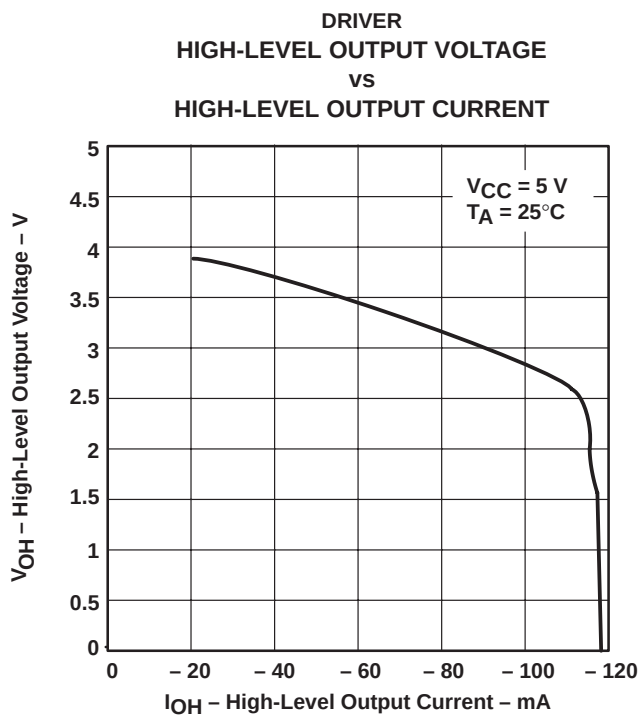


Figure 11

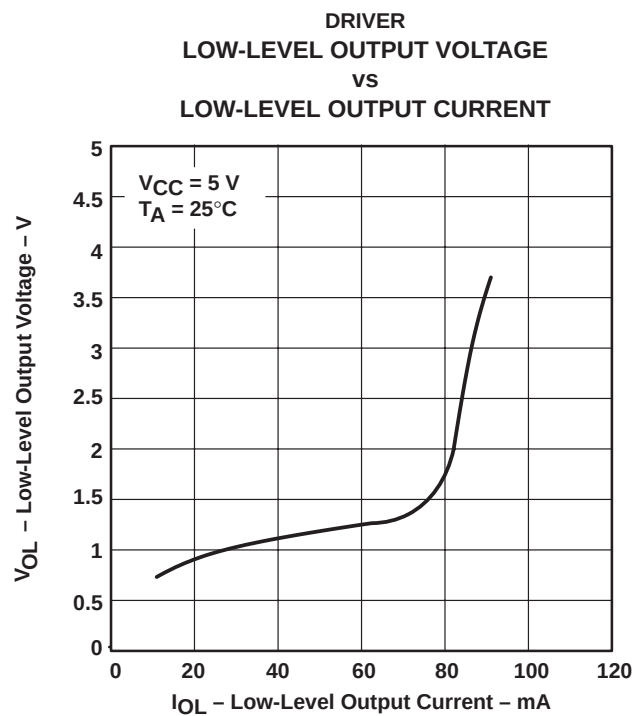


Figure 12

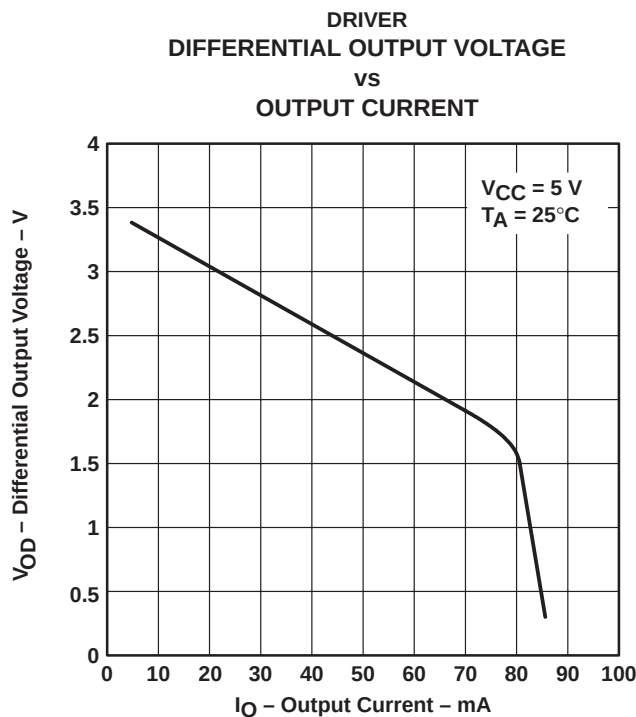


Figure 13

TYPICAL CHARACTERISTICS

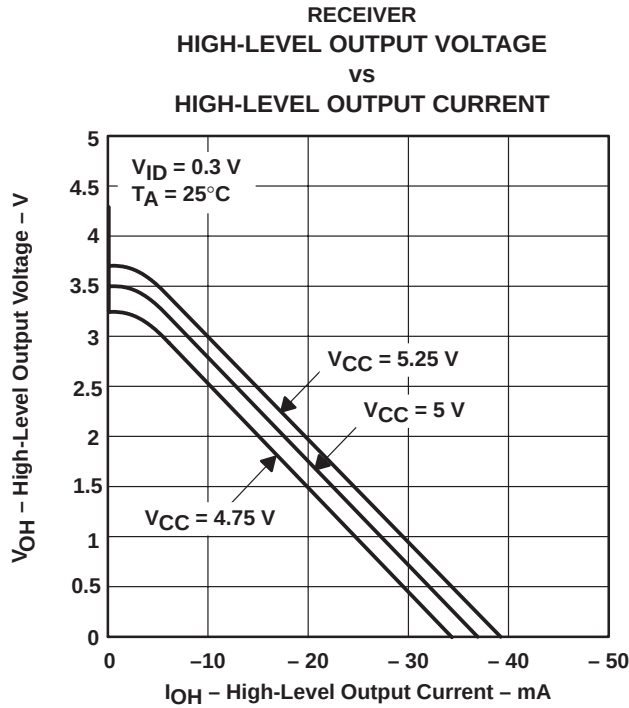


Figure 14

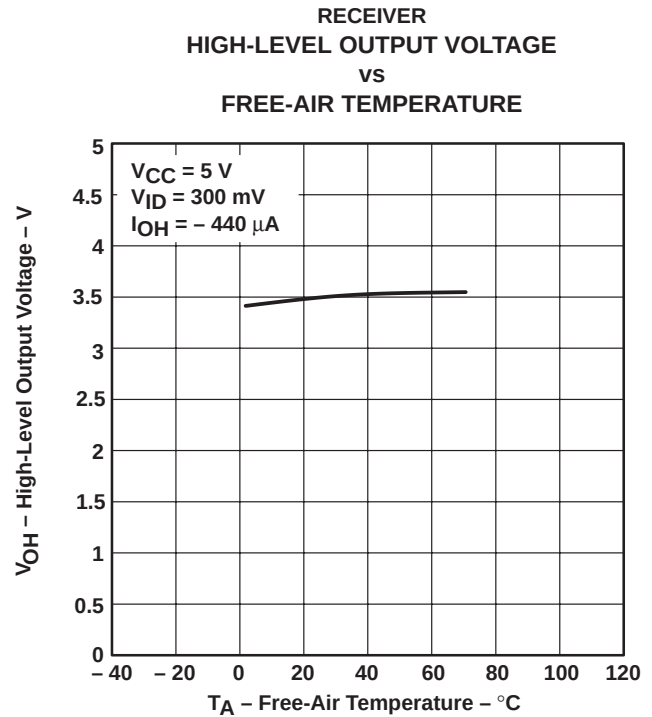


Figure 15

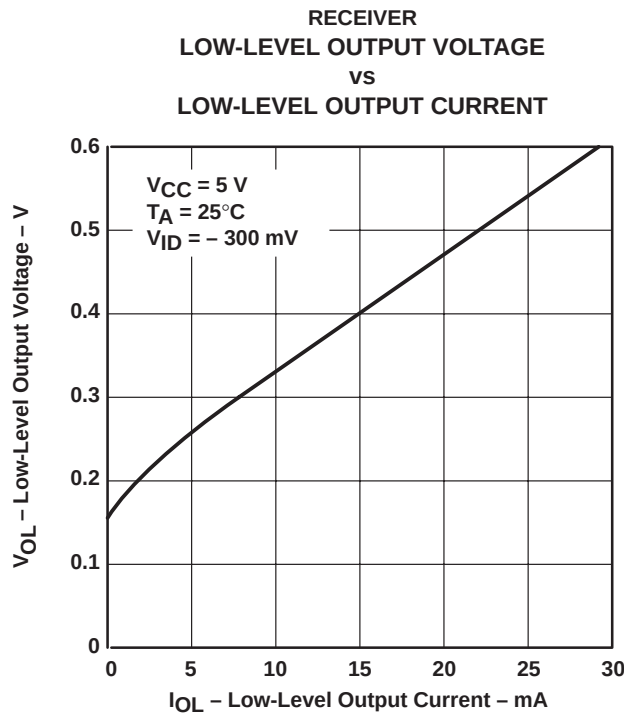


Figure 16

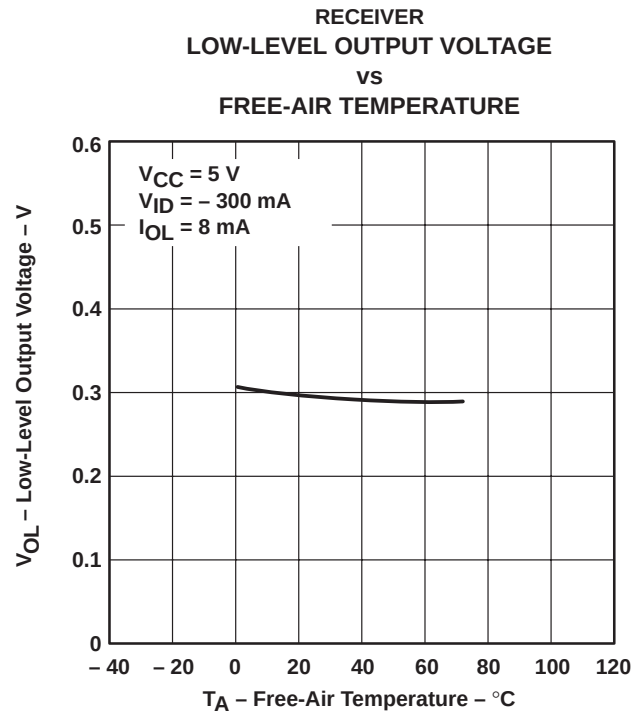


Figure 17

SN75ALS171, SN75ALS171A

TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

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TYPICAL CHARACTERISTICS

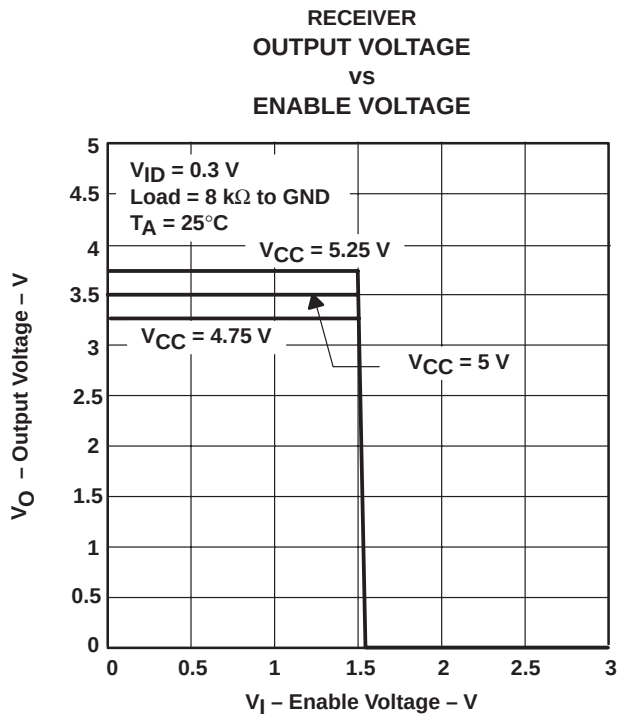


Figure 18

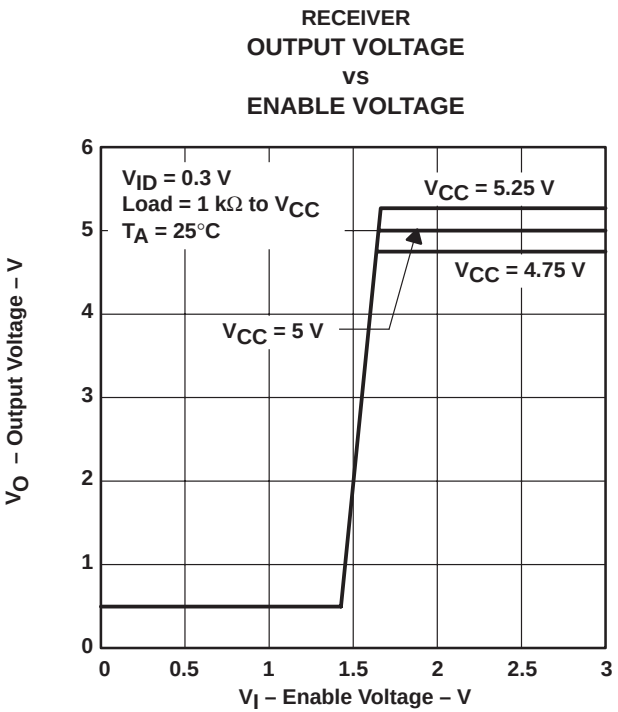
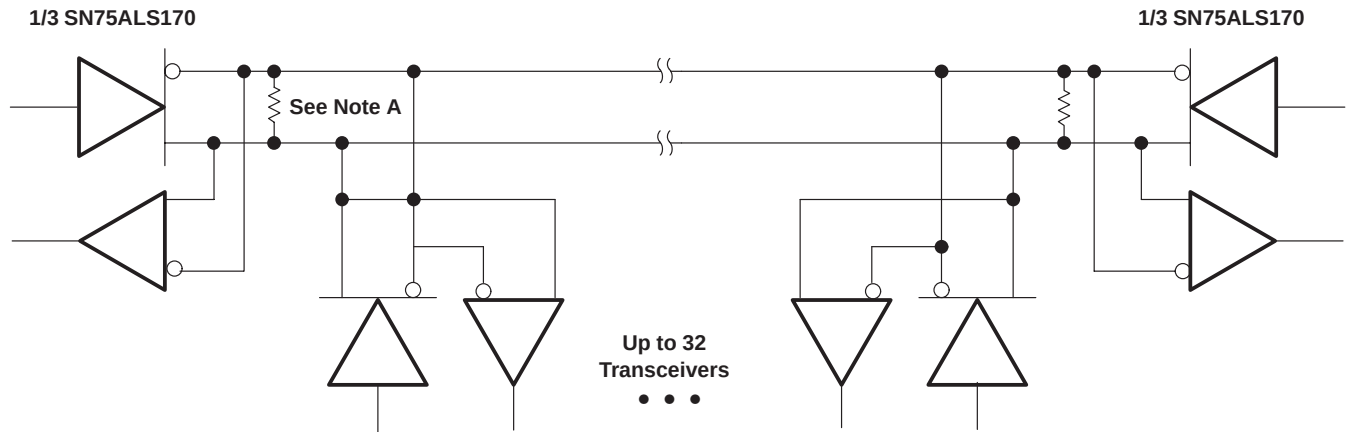


Figure 19

APPLICATION INFORMATION



NOTE A: The line should be terminated at both ends in its characteristic impedance. Stub lengths off the main line should be kept as short as possible.

Figure 20. Typical Application Circuit

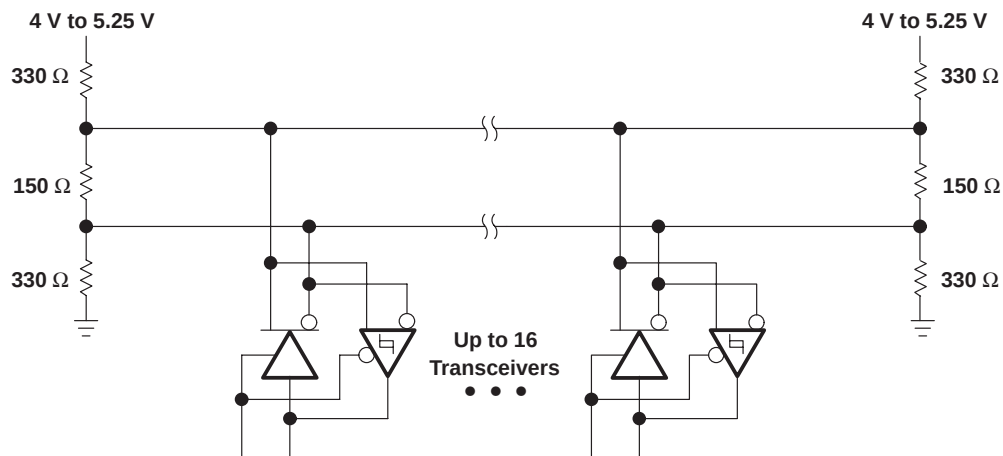


Figure 21. Typical Differential SCSI Application Circuit

SN75ALS171, SN75ALS171A TRIPLE DIFFERENTIAL BUS TRANSCEIVERS

SLLS056D – AUGUST 1987 – REVISED SEPTEMBER 1995

APPLICATION INFORMATION

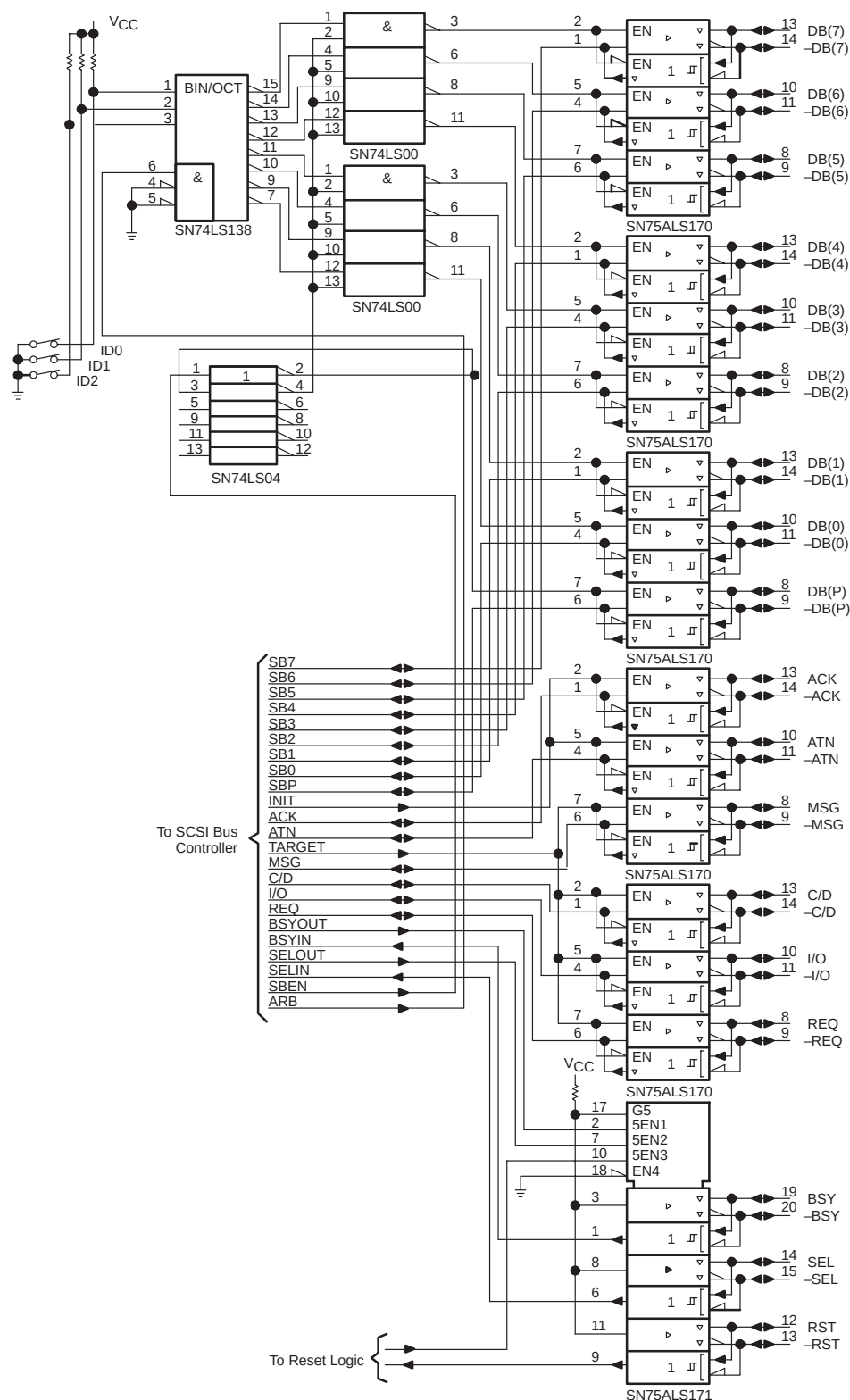


Figure 22. Typical Differential SCSI Bus Interface Implementation

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN75ALS171ADW	Active	Production	SOIC (DW) 20	25 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	75ALS171A
SN75ALS171ADW.A	Active	Production	SOIC (DW) 20	25 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	75ALS171A
SN75ALS171DW	Active	Production	SOIC (DW) 20	25 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	75ALS171
SN75ALS171DW.A	Active	Production	SOIC (DW) 20	25 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	75ALS171
SN75ALS171DWR	Active	Production	SOIC (DW) 20	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	75ALS171
SN75ALS171DWR.A	Active	Production	SOIC (DW) 20	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	75ALS171

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

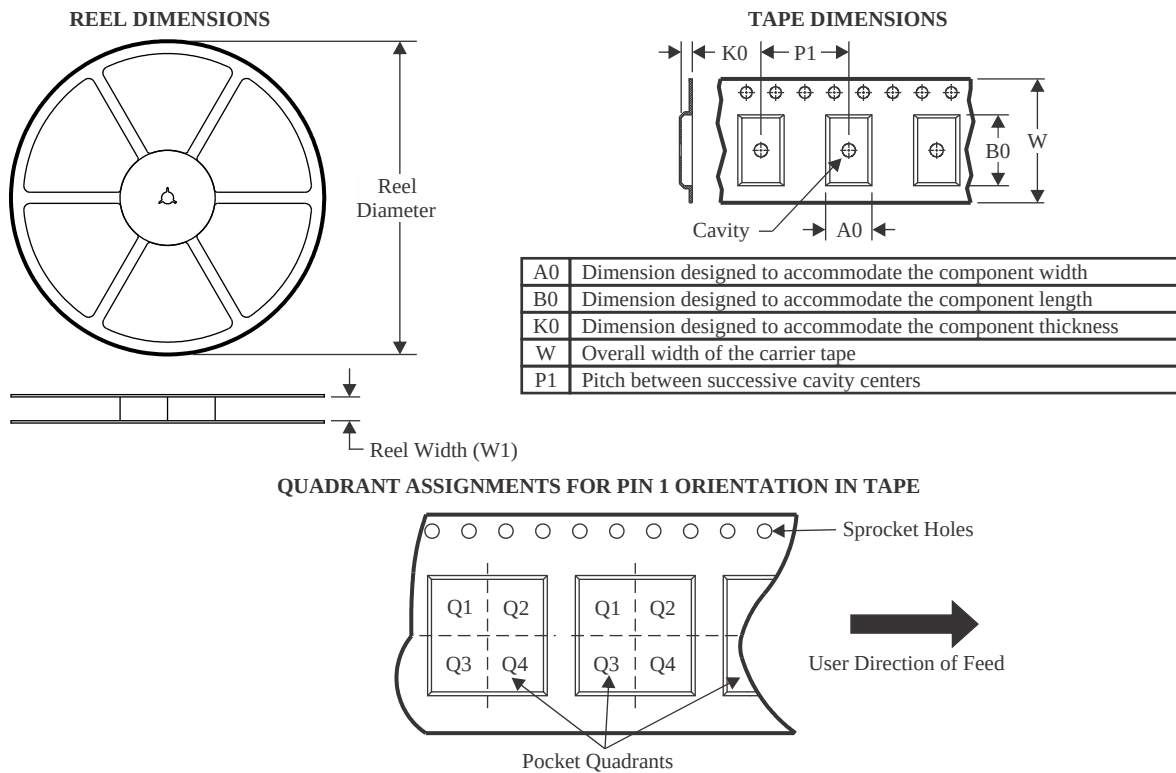
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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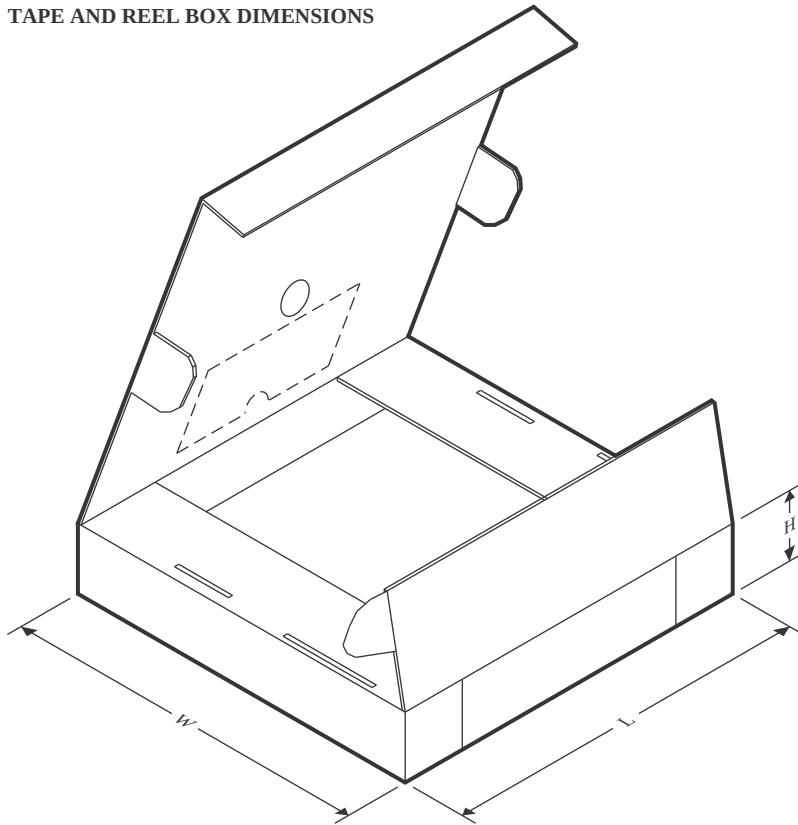
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN75ALS171DWR	SOIC	DW	20	2000	330.0	24.4	10.8	13.3	2.7	12.0	24.0	Q1

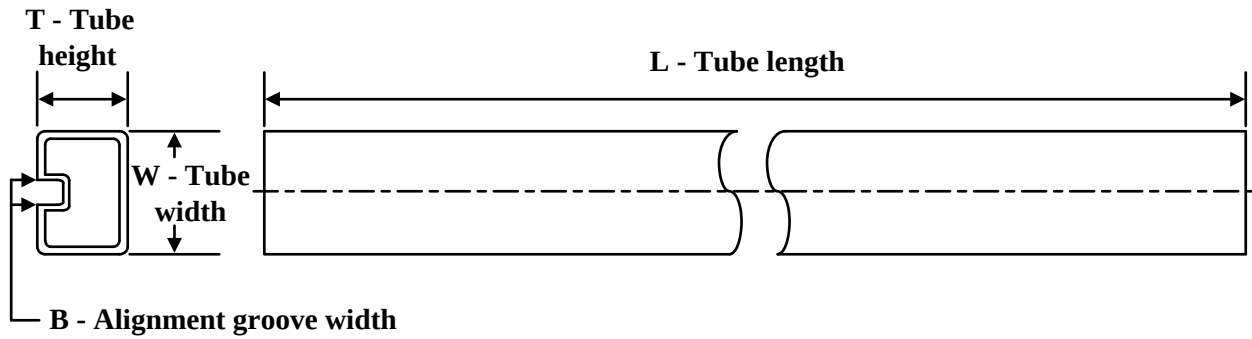
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

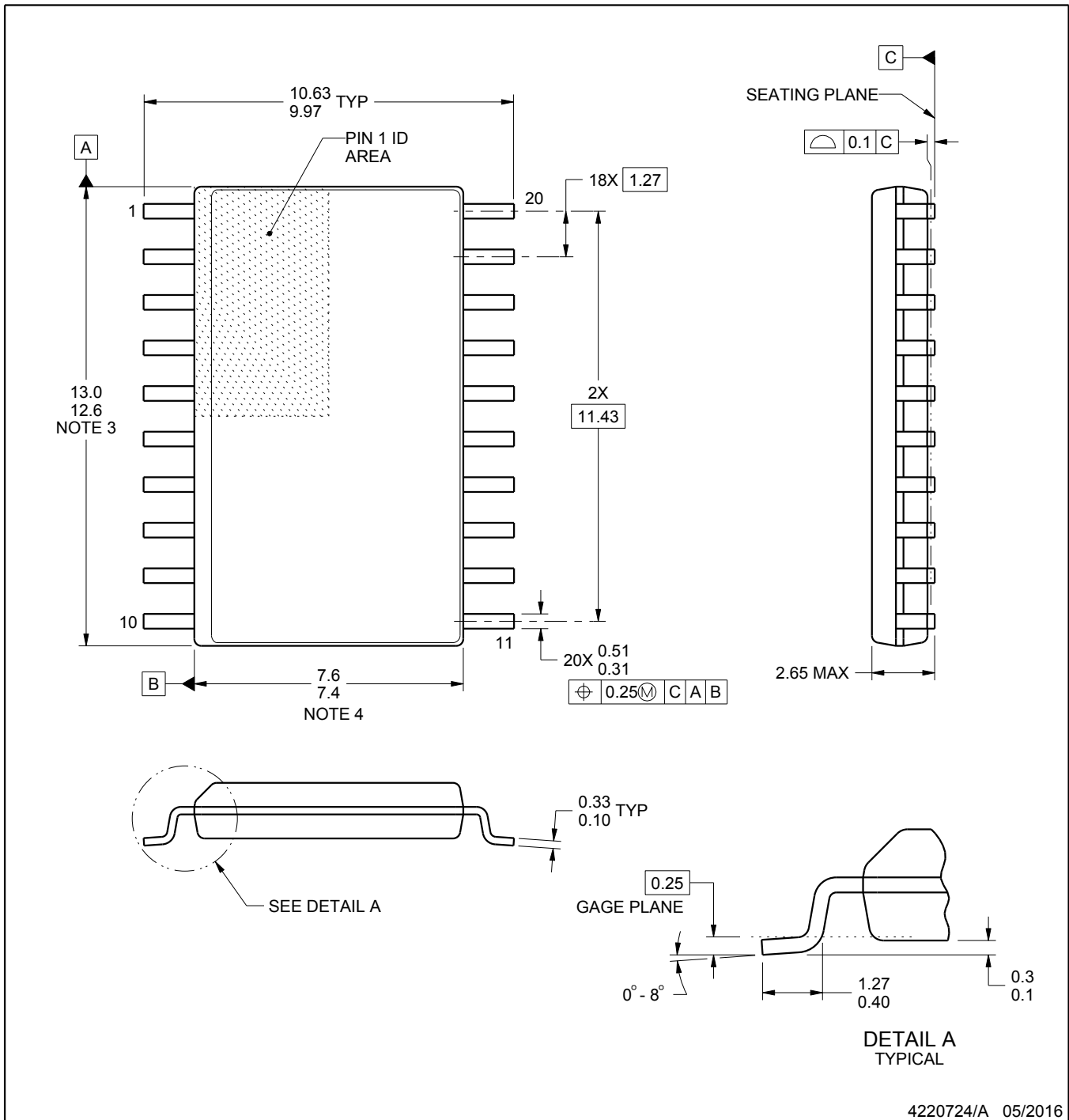
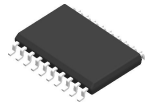
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN75ALS171DWR	SOIC	DW	20	2000	350.0	350.0	43.0

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
SN75ALS171ADW	DW	SOIC	20	25	506.98	12.7	4826	6.6
SN75ALS171ADW.A	DW	SOIC	20	25	506.98	12.7	4826	6.6
SN75ALS171DW	DW	SOIC	20	25	506.98	12.7	4826	6.6
SN75ALS171DW.A	DW	SOIC	20	25	506.98	12.7	4826	6.6



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NOTES:

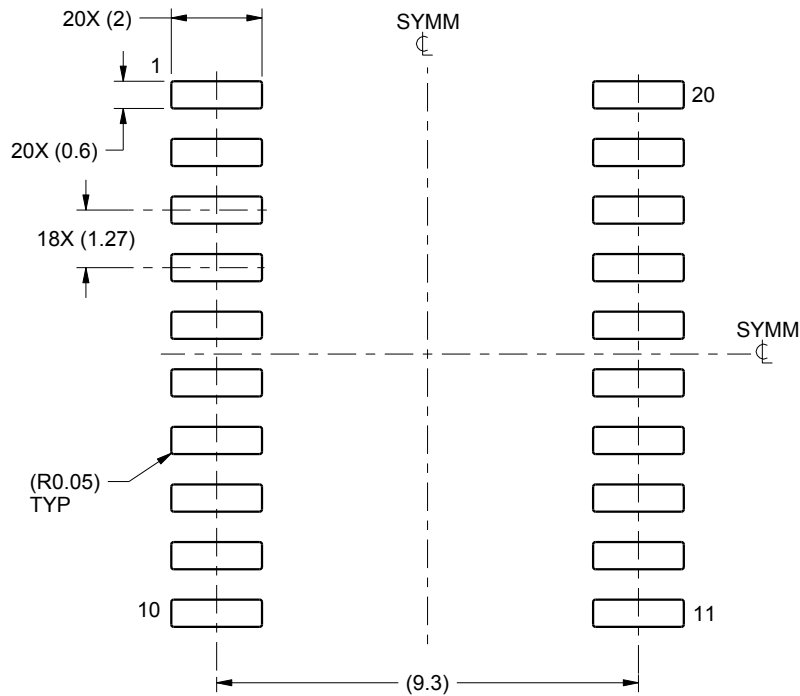
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm per side.
5. Reference JEDEC registration MS-013.

EXAMPLE BOARD LAYOUT

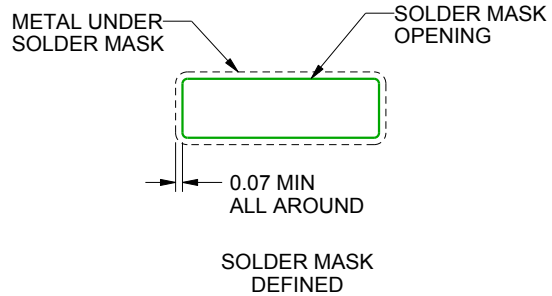
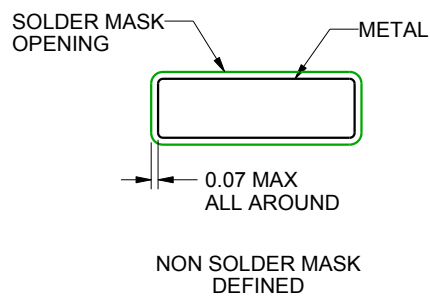
DW0020A

SOIC - 2.65 mm max height

SOIC



LAND PATTERN EXAMPLE
SCALE:6X



SOLDER MASK DETAILS

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

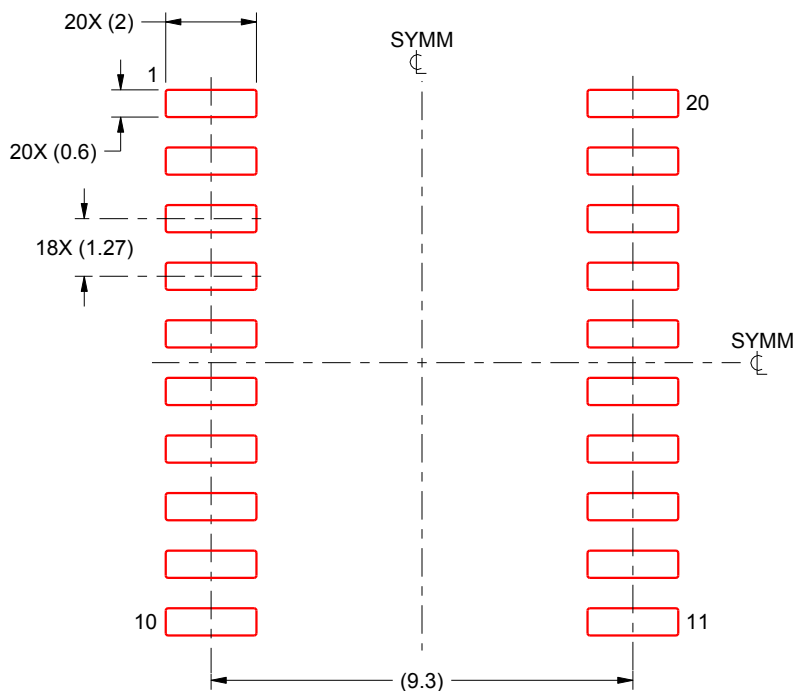
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DW0020A

SOIC - 2.65 mm max height

SOIC



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:6X

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NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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